

Komatsu Parts Procurement & SAP Business One Integration Specification

Operational Fleet Context, Komatsu PDX Manufacturer Ordering Rules, and Order Data Delivery Specification for ABS Consulting

TARGET PARTNER	PREPARED BY	VENDOR MASTER	DOCUMENT VERSION
ABS (SAP B1 Technical Team)	Motasem Ghanem (EQP Ops)	Komatsu Middle East (v000006)	v3.0 (Operational RFC & Data Manifest)

1. Operational Fleet Context: Mission-Critical Asset Management

Dar Al Hai General Trading operates a major heavy equipment fleet deployed across Kuwait's civil infrastructure, deep earthmoving, pipeline installation, and marine reclamation projects. The core operational fleet comprises heavy **Komatsu hydraulic excavators** (including PC500LC-10R and PC400-8R), high-tonnage crawler bulldozers (D375A-6), and heavy wheel loaders (WA470-6). In Kuwait's extreme operating conditions—where ambient summer temperatures consistently exceed 50°C and abrasive desert silica dust accelerates component wear—heavy earthmoving assets operate on continuous multi-shift production cycles. Machine uptime directly governs critical infrastructure contract milestones.

Maintenance and overhaul operations are centered at Dar Al Hai's **Shuwaikh Central Maintenance Workshop**, supported by the **Sulaibiya Logistics Staging Depot**. Genuine OEM replacement parts are procured directly from the regional manufacturer hub, **Komatsu Middle East FZE (KME)** in Dubai JAFZA, via their B2B digital portal (**Komatsu PDX**). In SAP Business One, Komatsu Middle East is registered as foreign Vendor **v000006** , transacting exclusively in **US Dollars (USD)**.

500,000+ KOMATSU PART MASTER CATALOG	100% Native AUTOMATED PRICING PASSTHROUGH	Zero Clerical DUAL DATA ENTRY INTO SAP B1
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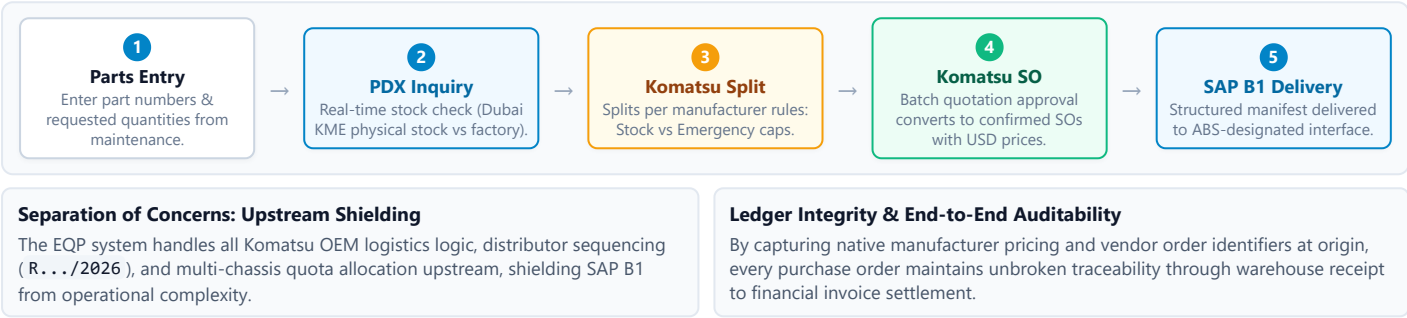
2. Guiding Integration Philosophy: Respecting Domain Boundaries

- Guiding Principle for ABS Consultants & Developers:**
ABS are the recognized SAP Business One implementation experts. We do not dictate SAP internal database schemas, table structures, or system triggers.
- Our objective is to present our operational reality, manufacturer constraints, and the clean data manifests produced by our EQP procurement engine, then collaborate with ABS to determine the optimal delivery mechanism:
- Native Manufacturer Pricing (Automatic Passthrough):** The unit price generated by Komatsu PDX on each confirmed Sales Order represents the final, binding price. For Emergency Orders (EO), Komatsu automatically embeds emergency rates directly into the line-item unit price. For Daily Stock (DS), it applies catalogue distributor rates.
 - Zero Surcharge Calculation in Middleware or SAP:** Neither SAP Business One nor the integration middleware calculates surcharges, markups, freight premiums, or custom pricing adjustments. SAP simply records the exact confirmed unit price certified by Komatsu.
 - Deterministic 1:1 Document Alignment:** Every confirmed Komatsu Sales Order (SO) corresponds to exactly one standard Purchase Order in SAP B1.
 - Upstream Manufacturer Isolation:** All Komatsu portal quirks, multi-chassis anti-hoarding limits, and sequential distributor references (R.../2026) are managed entirely inside EQP, shielding SAP B1 from operational complexity.

Operational Responsibility Matrix

DOMAIN AREA	EQP PROCUREMENT MIDDLEWARE (DAR AL HAI)	SAP BUSINESS ONE (ABS CONSULTING DOMAIN)
OEM Portal Logistics	Komatsu PDX live inventory check, anti-hoarding quotas, chassis binding.	Completely isolated from portal mechanics; receives verified order data.
Pricing & Surcharges	Captures binding USD unit prices with emergency premiums pre-baked.	Records exact certified unit prices; zero surcharge logic required in ERP.
Financial Ledger	Provides clean header & line manifests with unbroken audit references.	Authoritative book of accounts, inventory valuation, GRPO, and A/P matching.
System Governance	Manages quotation-to-SO lifecycle, retry queues, and idempotency locks.	Governs purchasing approvals, warehouse receipts, and vendor ledger postings.

3. End-to-End System Workflow Architecture



Summary for ABS Integration Engineers:

4. Operational Constraint: Why Orders Are Split on Komatsu PDX

In standard enterprise ERP configurations, ordering 50 units of a spare part is recorded as a single purchase requisition line. However, the **Komatsu Middle East PDX Portal** enforces strict manufacturer routing and anti-hoarding policies designed to safeguard factory inventory across the Middle East. Consequently, our EQP procurement middleware must partition requisitions into distinct sub-orders before submission.

Crucial Distinction for the ABS Consulting Team:

Order splitting is strictly an external Komatsu manufacturer policy constraint, NOT an internal SAP data modeling requirement. Sub-order generation and machine allocation exist solely to comply with Komatsu PDX portal validation rules prior to order submission. Once Komatsu approves and confirms the sub-orders, each Sales Order maps cleanly 1:1 into standard SAP Purchase Orders.

Manufacturer Allocation Logistics: Dubai JAFZA Hub vs. KLTD Japan Factory

Komatsu operates a two-tiered fulfillment strategy: fast-moving maintenance consumables are held at the regional distribution center in JAFZA (Dubai, UAE), whereas specialized mechanical, hydraulic, and structural assemblies are manufactured and held at Komatsu Ltd. (KLTD) factories in Japan. To ensure equitable parts distribution across GCC dealers and prevent speculative stockpiling, Komatsu strictly caps emergency order quantities per verified machine chassis serial.

Fulfillment Channels Comparison: Normal Daily Stock (DS/SO) vs. Emergency Breakdown (EO)

OPERATIONAL DIMENSION	NORMAL DAILY STOCK (DS / SO)	EMERGENCY BREAKDOWN (EO)
Physical Inventory Sourcing	Fulfilled directly from KME Dubai central distribution warehouse inventory.	Sourced from Komatsu Ltd. (KLTD) Japan factory or global emergency reserve.
Ordering Quotas & Caps	Uncapped bulk ordering. No maximum line-item limits imposed by Komatsu.	Strictly capped per chassis serial. Komatsu restricts maximum quantity per machine (e.g., max 20 units/chassis) to prevent distributor stockpiling.
Equipment & Asset Binding	No machine or customer required. Treated as general inventory replenishment.	Mandatory equipment binding. Must specify verified machine model (e.g. PC500LC-10R) and valid chassis serial (e.g. 100433).
Unit Pricing Mechanism	Base catalogue distributor rate, computed natively by Komatsu PDX.	Emergency rate, computed natively by Komatsu PDX directly on the unit price.
Distributor Reference	Sequential distributor reference number (e.g., R229/2026).	Sequential distributor reference numbers (e.g., R230/2026 , R231/2026).
Shipping & Logistics Mode	Consolidated sea/land freight to Shuwaikh Workshop receiving bay.	Expedited air freight directly to Kuwait International Airport clearance.
Target Delivery Due Date	DS: 10 Days (Dubai stock) • SO: 120 Days (Factory stock order)	EO: 30 Days (Emergency order turnaround)

5. Concrete Walkthrough: Requisition for 50 EA Fuel Filters (Part 600-311-6742)

When field maintenance requests 50 fuel filters, the EQP system checks live PDX inventory and automatically executes a hybrid split across channels:

Requisition: 50 EA • Fuel Filter (600-311-6742)				Komatsu PDX Status: KME Dubai Stock: 10 EA Factory Shortage: 40 EA		
SUB-ORDER	CHANNEL	SOURCE	ASSET BINDING	KOMATSU SO #	UNIT PRICE	SAP PO ACTION
R229/2026	DS Stock	KME Dubai (10 EA)	None (General Stock)	SO #0000278046	\$25.00 (Base)	Generates SAP PO #1
R230/2026	EO Emerg.	KLTD Japan (20 EA)	Excavator 100433	SO #0000278047	\$28.75 (EO Rate)	Generates SAP PO #2
R231/2026	EO Emerg.	KLTD Japan (20 EA)	Excavator 100434	SO #0000278048	\$28.75 (EO Rate)	Generates SAP PO #3

Pricing Passthrough Verification: Notice that the \$28.75 emergency rate is computed natively by Komatsu PDX and certified directly on the Sales Order. Neither EQP nor SAP B1 performs surcharge calculations.

6. Financial Reconciliation & Three-Way Matching Governance

Flawless Accounting & Warehouse Reconciliation in SAP Business One:

Each resulting SAP Purchase Order captures the exact Komatsu Sales Order Number and Distributor Reference in SAP PO reference fields. When Komatsu Middle East ships the goods, their shipping documents and commercial invoices correspond 1:1 with individual Komatsu SOs:

- Goods Receipt PO (GRPO):** Warehouse staff receive incoming parts in SAP B1 against the specific PO referencing Komatsu's shipping packing slip and delivery note.
- A/P Invoice 3-Way Matching:** Accounts Payable matches Komatsu's USD commercial invoice against the GRPO and PO with zero price variance, because SAP recorded the exact confirmed unit price certified by Komatsu.
- Currency Realization:** Foreign currency liability is recognized at transaction time, eliminating downstream ledger adjustments and manual credit memo reconciliations.

7. Operational Workflow Phase I: Part Entry & Fleet Allocation

Step 1: Part Entry, Real-Time Master Stock & Allocation

The maintenance engineer enters requested part numbers. The EQP engine verifies them against the Komatsu catalogue master, displays live inventory badges (**KME Stock** in Dubai, **EOR** emergency restriction, **KLTD** factory stock), and automatically calculates available quantities versus emergency shortage. The system also verifies part supersession numbers, ensuring superseded items are seamlessly replaced with current genuine factory specifications.

Genuine Verification
Validates part numbers instantly against the 500,000+ Komatsu catalogue master database.

Live Stock Partitioning
Distinguishes immediate Dubai JAFZA hub inventory from Japan factory lead-time items.

Supersession Control
Automatically flags superseded items and upgrades part numbers to current active revisions.

1 Requested Parts & Item Types

Add multiple item numbers. The automated dispatcher bundles line items together into unified quotations up to the sub-order max.

Import / Paste

Verify Parts

+ Add Part Item

PART NUMBER	DESCRIPTION	REQUESTED QTY	MAX / SUB-ORDER	UNIT PRICE (\$)	LINE TOTAL (\$)
2A8-62-12230	HOSE	12	2	66.310	\$795.72
✓ Verified Genuine (PC500LC-10)					
2A8-62-11751	HOSE	10	1	70.230	\$702.30
✓ Verified Genuine (PC500LC-10)					
Total Summary: 2 Item Types		22 EA		Est. Total:	\$1498.020

Clear

Figure 1: Part Entry table showing catalogue verification (Verified Genuine PC500LC-10), unit prices, and line-item quantity breakdowns.

Step 2: Target Fleet & Multi-SN Allocation (Emergency Orders Only)

When items route through the Emergency breakdown channel, the dispatcher selects the target customer account (e.g. *Laala Al-Kuwait Real Estate Co.*) and selects compatible heavy equipment chassis serials (e.g. *PC500LC-10R* pool). Active equipment pools are filtered by machine model and maintenance schedule. For 100% Stock orders, machine allocation is completely bypassed.

Active Fleet Pools
Filters compatible machinery by customer contract, model code, and active jobsite location.

Quota Balancing
Enforces Komatsu's strict chassis serial caps to prevent anti-hoarding portal submission errors.

Continuous Audit Keys
Maintains uninterrupted distributor reference sequencing (Starting DB Order No: R229/2026).

2 Target Fleet & Multi-SNs Allocation

Select target equipment serial numbers to distribute inquiry quotations

Customer Account

Machine Type

Model (Compatibility)

Laala Al-Kuwait Real Estate

All Types (84)

★ PC500LC-10R (100% Mat)

Serial Numbers in Pool: 20

★ Auto-Select Compatible

Select All (20)

Clear

Custom SNs Input

100461 (PC500LC-10R)

Full Match

100463 (PC500LC-10R)

Full Match

100466 (PC500LC-10R)

Full Match

100468 (PC500LC-10R)

Full Match

100477 (PC500LC-10R)

Full Match

100478 (PC500LC-10R)

Full Match

100480 (PC500LC-10R)

Full Match

100504 (PC500LC-10R)

Full Match

100510 (PC500LC-10R)

Full Match

100520 (PC500LC-10R)

Full Match

3 Order Sequence & PDX Settings

Starting order sequence, comments, and submission mode

Starting DB Order No *

R229/2026

Sequential reference number

Order Comments / PO Reference

Urgent Breakdown

Simulation Mode (Dry Run)

Calculate unified quotations and sub-orders without dispatching live records to Komatsu PDX.

Generated Sub-Orders:

10 Quotation(s)

Target Machine Pool:

20 SNs allocated

Est. Total Order Value:

\$1498.020

Figure 2: Active Machine Pool Selection (filtered for compatible equipment) and continuous distributor reference setup (Starting DB Order No: R229/2026).

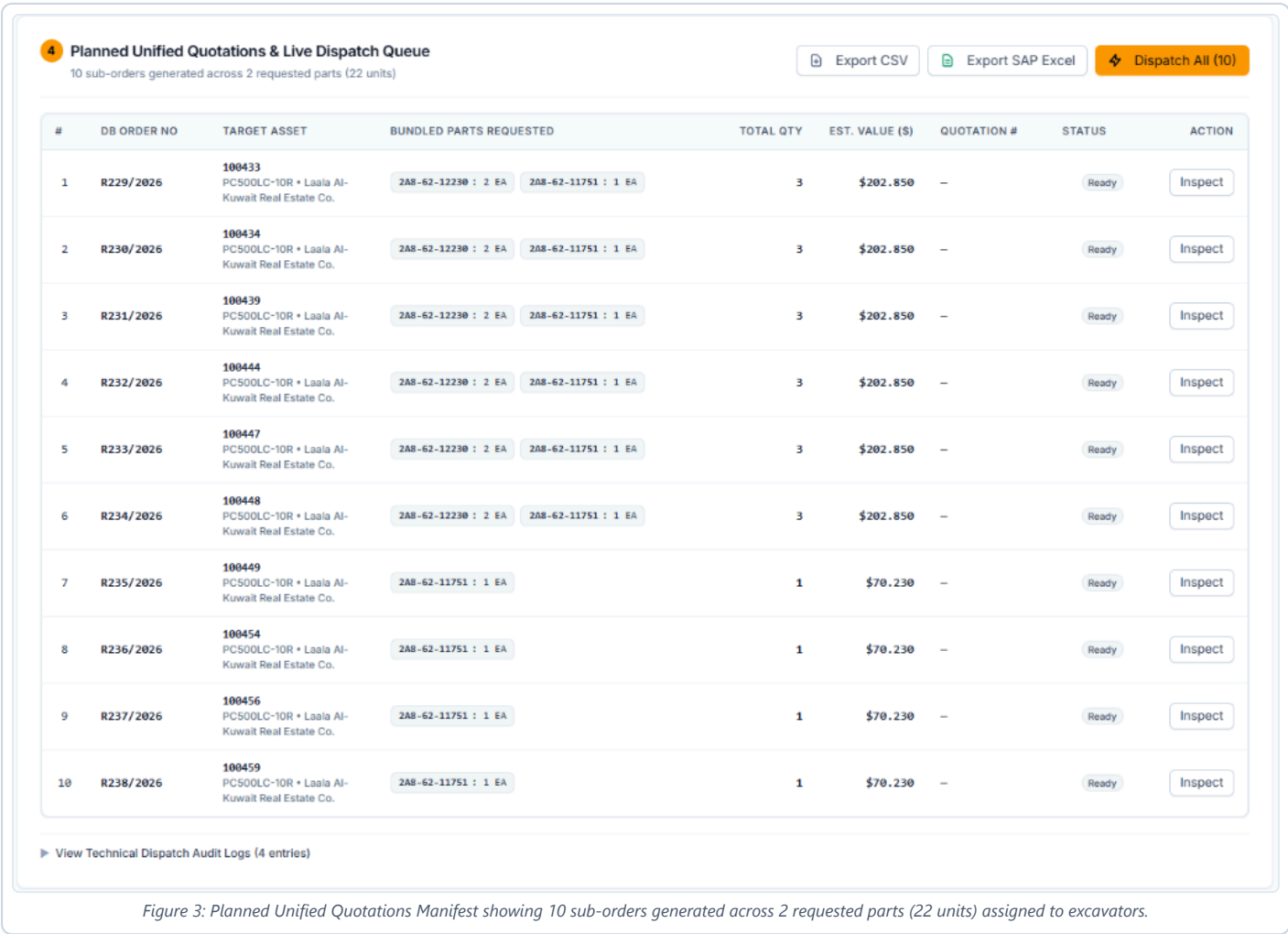
Automated Dispatch Preparation & Dry-Run Simulation:

The system calculates estimated quotation counts and order values in real-time, allowing operators to run dry-run simulations before dispatching live records to Komatsu PDX. This guarantees full data validation prior to external transmission, eliminating rejected orders and ensuring continuous distributor audit numbering.

8. Operational Workflow Phase II: Planned Unified Quotations Manifest

Step 3: Planned Unified Quotations Manifest & Live Dispatch Queue

The EQP splitting engine bundles line items across assigned excavator chassis into unified distributor quotations within Komatsu sub-order caps. Each sub-order receives a sequential distributor reference number (R229/2026 through R238/2026) ensuring continuous audit numbering. This bundling optimizes freight efficiency and minimizes paperwork overhead.



Queue Structure & Dispatch Governance Controls

QUEUE ATTRIBUTE	OPERATIONAL IMPLEMENTATION	GOVERNANCE & AUDIT FUNCTIONALITY
DB Order Number	Sequential format: R<Seq>/2026	Ensures non-gapping, continuous audit tracking across distributor submissions.
Target Asset Binding	Chassis SN + Model + Customer Account	Enforces Komatsu's manufacturer requirement for emergency breakdown allocation.
Bundled Part Lines	Multi-part bundling per quote (e.g. 2 Hoses)	Maximizes shipping efficiency and minimizes individual sub-order transaction count.
Export & Audit Tools	CSV Export • SAP Excel Export • Audit Logs	Provides immediate offline backup and full dispatch traceability for procurement supervisors.

Dry-Run Pre-Flight Validation

Simulates entire submission payloads to confirm chassis compatibility and quota adherence before transmitting to Komatsu Middle East.

Automated Re-Packaging Logic

If Komatsu adjusts line-item allocations, EQP re-packages affected orders seamlessly while maintaining distributor reference integrity.

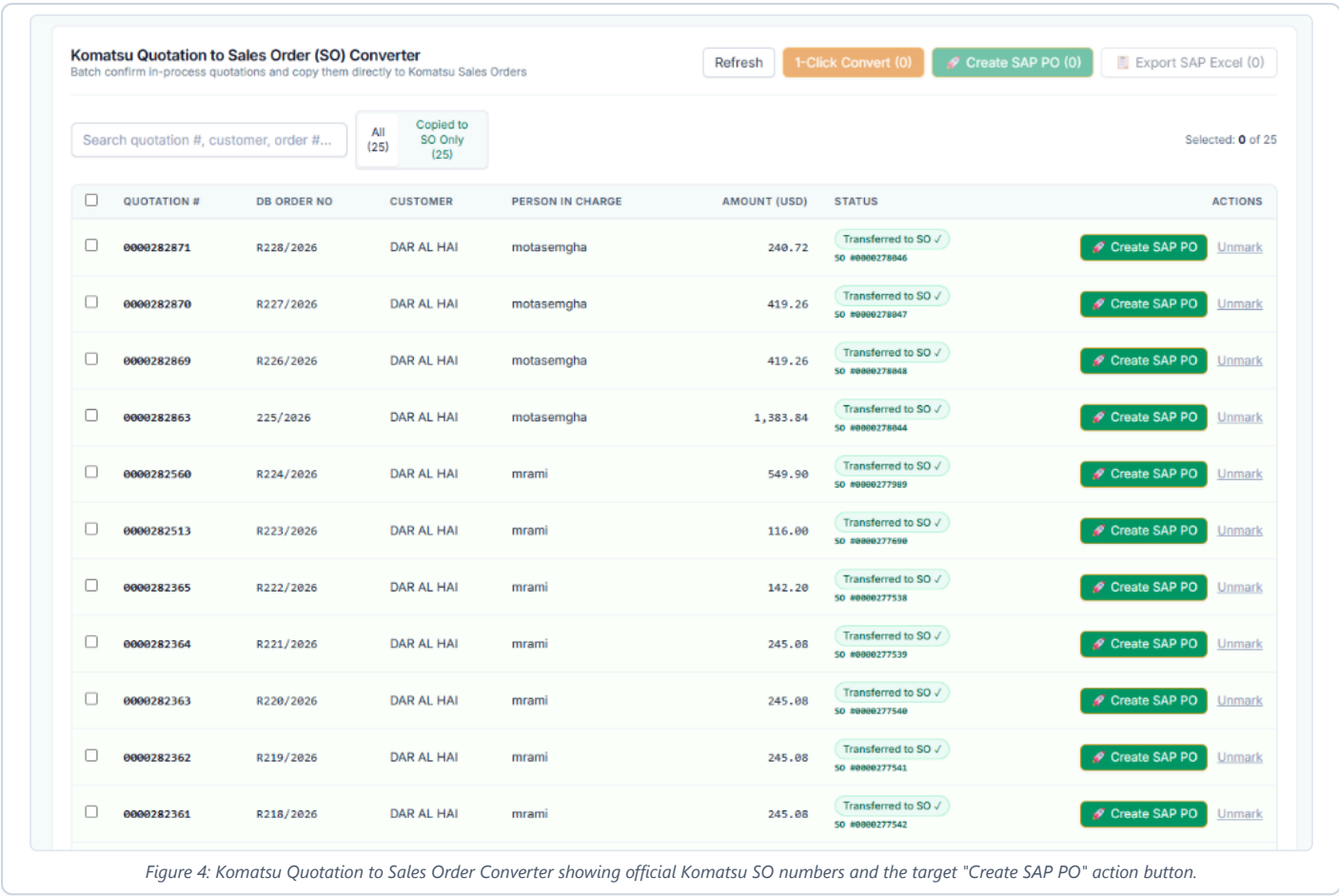
Batch Dispatch Resilience:

The live dispatch queue provides granular inspection of each sub-order prior to execution, preventing accidental submissions and ensuring strict compliance with Komatsu Middle East ordering quotas.

9. Operational Workflow Phase III: Quotation-to-SO Conversion & SAP Trigger

Step 4: Komatsu Quotation to Sales Order (SO) Converter

Once submitted to Komatsu PDX, the portal confirms the sub-orders with official Komatsu Quotation Numbers (e.g. 0000282871). The EQP procurement interface enables 1-Click batch conversion to certified **Komatsu Sales Orders (SO)** (e.g. SO #0000278046) carrying binding USD purchase prices.



The SAP Order Delivery Execution Lifecycle

Order Delivery & Confirmation Lifecycle:

At this stage, the Komatsu Sales Order is officially confirmed with certified USD pricing. Clicking "Create SAP PO" initiates the following automated sequence:

1. Manifest Serialization:

EQP middleware packages confirmed part numbers, quantities, certified unit prices, and tracking references into a structured order manifest.

2. Transmission to SAP B1:

Data is dispatched to the integration interface defined and configured by ABS.

3. Response & Persistence:

Upon successful document creation in SAP, the returned SAP PO document number is recorded in the EQP database for complete auditability.

4. Idempotency Lock:

The button updates to "SAP PO #[DocNum] ✓" and locks permanently, eliminating duplicate document creation.

Integration State Machine & Governance Statuses

SYSTEM STATE	UI INDICATOR	OPERATIONAL BEHAVIOR & SAFEGUARDS
PDX_CONFIRMED	SO Confirmed	Official Komatsu Sales Order confirmed with binding USD prices; ready for delivery to SAP.
TRANSMITTING	Transmitting...	Payload dispatched to ABS interface; retry lock engaged to prevent duplicate concurrent submissions.
COMMITTED_IN_SAP	SAP PO #Confirmed ✓	PO confirmed by SAP B1; official DocNum stored permanently; UI locked against duplicate posting.
RETRY_QUEUED	Retry Pending	Temporary network timeout captured; payload safely staged for one-click operator retry without data corruption.

Enterprise Idempotency Protection & Error Handling:

EQP utilizes unique idempotency hashing combining Vendor Code, Komatsu SO Number, and Distributor Reference. In the event of transient network failure or SAP endpoint unavailability, subsequent attempts cannot produce duplicate documents in SAP Business One.

10. EQP Order Data Manifest: Structured Data Delivered to SAP B1

For every confirmed Komatsu Sales Order, our EQP procurement engine produces a clean, validated, and complete order manifest. We do not prescribe internal SAP table structures or dictate database fields; rather, we provide our exact operational business data below so that ABS consultants can determine the optimal mapping and ingestion logic in SAP Business One.

Automatic Pricing Passthrough Guarantee:
All line-item unit prices supplied in this manifest are final, legally binding manufacturer prices certified directly by Komatsu Middle East.
Surcharges (whether emergency breakdown premiums or standard catalogue rates) are already fully incorporated into the unit price by Komatsu PDX. No pricing recalculation, surcharge calculation, or freight markup logic is needed in the integration middleware or inside SAP B1.

A. Order Header Data Manifest

BUSINESS FIELD	DATA TYPE	OPERATIONAL CONTEXT & PURPOSE	EMERGENCY ORDER	STOCK ORDER
Vendor Code	String	Komatsu Middle East vendor account in SAP	"V000006"	"V000006"
Komatsu SO No.	String	Official confirmed Sales Order issued by Komatsu PDX	"0000278047"	"0000278046"
Distributor Ref No.	String	Sequential distributor reference tracking number	"R230/2026"	"R229/2026"
Komatsu Quote No.	String	Preceding quote reference issued by Komatsu portal	"0000282872"	"0000282871"
Fulfillment Channel	String	Procurement channel (EO vs DS)	"EMERGENCY"	"STOCK"
Order Date / Due Date	Date (ISO)	Order placement date & target delivery date	2026-10-05 → 2026-11-04 (30d)	2026-10-05 → 2026-10-15 (10d) (SO: +120d → 2027-02-02)
Currency	String (ISO)	Billing currency for Komatsu Middle East	"USD"	"USD"
Equipment Serial	String / null	Machine chassis serial (Mandatory for EO; null for Stock)	"100433"	null (Stock)
Equipment Model	String / null	Equipment model code (e.g. PC500LC-10R)	"PC500LC-10R"	null (Stock)
Document Remarks	String	Distributor reference tracking number	"R230/2026"	"R229/2026"

B. Line Items Data Manifest

LINE	OEM PART NUMBER	PART DESCRIPTION	QTY	UNIT PRICE (USD)	LINE TOTAL	SUGGESTED WHS
1	600-311-6742	FUEL FILTER	20.00	\$28.75 (Certified EO)	\$575.00	003 (Optional)
2	07000-12015	O-RING	5.00	\$4.20 (Certified EO)	\$21.00	003 (Optional)

C. Clean Order Manifest Structure (Sample JSON Provided by EQP System)

```
{
  "orderHeader": {
    "vendorCode": "V000006",
    "komatsuSalesOrderNo": "0000278047",
    "distributorRefNo": "R230/2026",
    "komatsuQuotationNo": "0000282872",
    "fulfillmentChannel": "EMERGENCY_ORDER",
    "orderDate": "2026-10-05",
    "deliveryDueDate": "2026-11-04",
    "currency": "USD",
    "targetEquipmentSerial": "100433",
    "targetEquipmentModel": "PC500LC-10R",
    "comments": "R230/2026"
  },
  "lineItems": [
    {
      "lineNumber": 1,
      "partNumber": "600-311-6742",
      "description": "FUEL FILTER",
      "quantity": 20.0,
      "unitPriceUSD": 28.75,
      "lineTotalUSD": 575.00,
      "suggestedWarehouse": "003",
      "suggestedTaxCode": "P0"
    }
  ]
}
```

Note on Warehouse and Tax Defaults: Warehouse (003 - Shuwaikh Central Workshop) and Tax Code (P0 - 0% Import) can either be delivered in payload or determined natively by SAP B1 Business Partner defaulting rules, per ABS's guidance.

11. Consultative Technical Request for ABS Consulting (RFC)

ABS brings specialized domain expertise in Dar Al Hai's SAP Business One configuration, financial workflows, and database governance. We respect your architectural ownership of the SAP environment and invite your team to define how our EQP procurement engine should interface with SAP B1. Please review the following technical points and advise on your preferred approach:

1. Preferred Ingestion Architecture & Communication Interface

How would ABS prefer our system to deliver confirmed order manifests to SAP B1?

- **Option A:** Standard SAP B1 Service Layer REST API (direct HTTP POST from EQP backend).
- **Option B:** Dedicated integration staging table with an automated SAP B1 transactional service.
- **Option C:** ABS custom API connector or intermediate middleware gateway.

2. Preferred Payload Schema & Reference Field Mapping

Does ABS prefer our system to format payloads directly into a specific SAP B1 document structure, or would you prefer our clean JSON manifest shown on Page 6? Where should our tracking references (Komatsu SO #, Distributor Ref R.../2026, and Machine Chassis Serial) be stored in SAP Purchase Orders (e.g. BP Reference No., Remarks, or custom User-Defined Fields)?

3. Item Master Onboarding Preference (Handling New Komatsu Part Numbers)

With over 500,000 Komatsu parts in the catalogue, occasionally a requested part is not yet registered in SAP B1. What is ABS's preferred procedure when an incoming order line references an unregistered part number?

- **Recommendation A:** Provide an Item Master onboarding endpoint / capability allowing EQP to pre-create missing part numbers using verified Komatsu catalogue descriptions and item groups prior to PO generation.
- **Recommendation B:** Map unregistered items to a designated non-stock / service item code with explicit line text descriptions.
- **Recommendation C:** Queue unlinked items for purchasing officer review before PO creation.

4. Document Status Workflow (Approved Open PO vs. PO Draft)

Should the automated integration create open, active Purchase Orders directly, or should it create Purchase Order Drafts for review and manual posting during the initial pilot phase?

5. Dedicated API Service User & Authentication Protocol

What authentication protocol does ABS recommend for our automated service (e.g. dedicated API service user credentials, session cookies, OAuth tokens)? We request a dedicated service user with scoped permissions to avoid dependence on individual employee accounts.

6. Sandbox Connectivity & Access Provisioning

Please provide sandbox/test environment access details (endpoint URL, test company database name, credentials, and firewall whitelisting requirements) so that connectivity and schema validation can begin.

ABS Technical Configuration Matrix

Integration Parameter	Responsible Party	Suggested Baseline	ABS Specification
Ingestion Interface	ABS Consulting	Service Layer / Staging / API	[To be specified by ABS]
Endpoint URL & Port	ABS Consulting	https://sap.daralhai.com: [port]/...	[To be provided by ABS]
Dedicated API User	ABS Consulting	Scoped service account	[To be provisioned by ABS]
Item Master Policy	Joint Technical Decision	API Auto-creation (Rec. A)	[ABS preferred method]
Document Status	Dar Al Hai Ops & ABS	PO Draft (Pilot) → Open PO	[To be confirmed by ABS]